# Econometrics Midterm

Multiple Choice · Econometrics 1–3 (Slides), Wooldridge Ch.1–6, Quiz 2 · 54 questions

1. An econometrician wants to learn the average hourly wage of all workers in Korea and collects data on 2,000 randomly selected workers. Which statement is correct?
   a. A sample is used only when the population is hypothetical
   b. The 2,000 workers are the population, because they are the units actually analyzed
   c. The 2,000 workers are the sample, and the estimates are statements about what the population of all Korean workers is like
   d. The estimates describe only the 2,000 workers and say nothing about other workers

2. Which statement best defines external validity?
   a. Whether the OLS estimator is unbiased within the sample used
   b. Whether the sample size is large enough to compute standard errors
   c. Whether the R-squared of the regression is high
   d. The validity of applying the conclusions of a study outside the context of that study

3. A researcher surveys only students at one private university in Seoul to estimate the average monthly smartphone spending of all Korean adults. What is the main concern?
   a. OLS cannot be computed because the population is too large
   b. The sample is unlikely to be representative of the population, so the conclusions may have weak external validity
   c. The estimates are automatically unbiased for all Korean adults because the students were chosen at random within the university
   d. The population becomes hypothetical, so no estimate is possible

4. Consider the model $\log(\text{price}) = \beta_0 + \beta_1 \text{subsidy} + \beta_2 \log(\text{sunlight}) + u$. How should $\beta_1$ be interpreted?
   a. The elasticity of price with respect to subsidy
   b. The slope from a simple regression of price on subsidy
   c. The change in log(price) from a one-unit increase in subsidy, holding log(sunlight) fixed (a partial slope)
   d. The total change in price from a one-unit increase in subsidy, including any change in sunlight it causes

5. In the model $\log(\text{price}) = \beta_0 + \beta_1 \text{subsidy} + \beta_2 \log(\text{sunlight}) + u$, which statement about subsidy is correct?
   a. Subsidy has a linear relationship with price, but not with log(price)
   b. Subsidy has a linear relationship with log(price), but not with price
   c. Subsidy has a linear relationship with both price and log(price)
   d. Subsidy has a linear relationship with neither price nor log(price)

6. In the model log(price) = $\beta_0 + \beta_1$ subsidy + $\beta_2$ log(sunlight) + u, which statement about sunlight is correct?

a. $\beta_2$ is the % change in price for a one-unit increase in sunlight

b. $\beta_2$ is the change in log(price) for a 1% increase in price

c. log(sunlight) has a linear relationship with log(price), so $\beta_2$ is approximately the % change in price for a 1% increase in sunlight, holding subsidy fixed

d. Sunlight has a linear relationship with price, so $\beta_2$ is the change in price for one more unit of sunlight

7. Suppose log(price) = $\beta_0 + \beta_1$ battery + $\beta_2$ battery$^2$ + u with $\beta_2 \neq 0$. Which statement is correct?

a. $\beta_1$ alone is the constant slope of battery for every value of battery

b. The model violates the assumption of linearity in parameters, so OLS cannot be used

c. Battery has a linear relationship with log(price) because the model is linear in parameters

d. Battery does not have a linear relationship with log(price); the effect of one more unit of battery depends on the current level of battery

8. Suppose log(price) = $\beta_0 + \beta_1$ battery + $\beta_2$ sunlight + $\beta_3$ battery× sunlight + u with $\beta_3 \neq 0$. Holding sunlight fixed, what is the effect on log(price) of one more unit of battery?

a. $\beta_3$· battery

b. $\beta_1$

c. $\beta_1 + \beta_3$

d. $\beta_1 + \beta_3$· sunlight

9. Which statement describes random sampling?

a. Every element has an equal probability of being assigned to the treatment or control group

b. The researcher chooses units that look typical of the population

c. Every element in the population has an equal probability of being selected into the sample

d. The sample contains more than half of the population

10. A researcher has a random sample of 500 workers and then flips a coin for each worker to decide who joins a job training program. The coin flip is an example of:

a. Random sampling

b. Perfect collinearity

c. Partialling out

d. Random assignment

11. Consider Model 2: wage = $\beta_0 + \beta_1$ educ + u, where ability is not included. People with better ability tend to get more education. Why can the OLS estimate of $\beta_1$ be biased even with a random sample?

a. There is no sample variation in educ

b. Ability goes into u and is correlated with educ, so the zero conditional mean condition E(u|educ) = 0 fails

c. Random sampling guarantees unbiasedness, so the estimate cannot be biased

d. The homoskedasticity assumption fails, which always causes bias

12. In Model 2, wage = $\beta_0 + \beta_1$ educ + u (ability omitted), suppose ability raises wage and is positively correlated with education. The OLS estimate of $\beta_1$ tends to:

a. Have a bias whose sign can never be determined
b. Understate the true return to education (downward bias)
c. Be unbiased, because ability is unobserved
d. Overstate the true return to education (upward bias)

13. The slides state: "OLS requires random sampling. It may also require random assignment." When estimating the effect of education on wages with Model 2 (wage = $\beta_0 + \beta_1$ educ + u), why would random assignment help?

a. Random assignment removes the need for any error term
b. Random assignment increases the R-squared of the regression
c. If educ were randomly assigned, it would be unrelated to ability and other factors in u, so the zero conditional mean condition would hold
d. Random assignment makes the sample representative of the population

14. A data set records the crime rate and the number of police officers for the same 150 cities in both 2020 and 2023. What type of data is this?

a. Pooled cross sections
b. Cross-sectional data
c. Panel (longitudinal) data
d. Time series data

15. A researcher draws a random sample of house prices in 1993 and a new, independent random sample of house prices in 1995 to evaluate a property tax change. What type of data is this?

a. Panel data
b. A single cross section
c. Time series data
d. Pooled cross sections

16. Which feature is typical of time series data but NOT of a cross section drawn by random sampling?

a. Observations are typically serially correlated, and their ordering conveys important information
b. Observations are more or less independent of each other
c. The ordering of observations does not matter
d. All observations are collected at the same point in time

17. In the simple regression y = $\beta_0 + \beta_1$ x + u, the OLS estimates $\hat{\beta}_0$ and $\hat{\beta}_1$ are chosen to:

a. Minimize the sum of squared residuals
b. Make the fitted value equal to y for every observation
c. Minimize the sum of residuals
d. Minimize the sum of absolute residuals

18. In a simple regression of y on x, $R^2 = 0.25$. Which statement is correct?
    a. 25% of the observations lie exactly on the regression line
    b. 25% of the sample variation in y is explained by x
    c. x causes 25% of the change in y
    d. The estimated slope equals 0.25

19. The fitted regression is log(wage) = 0.584 + 0.083educ. How should 0.083 be interpreted?
    a. A 1% increase in education raises the wage by 8.3%
    b. One more year of education is associated with an approximately 8.3% higher wage
    c. One more year of education raises the wage by 0.083%
    d. One more year of education raises the wage by $0.083

20. Real earnings (in $1,000s) are regressed on train (= 1 if the person joined a randomly assigned job training program, 0 otherwise): earn = 4.55 + 1.79train. What does 1.79 measure?
    a. Nothing, because OLS cannot be used when x is binary
    b. Participants earn about 1.79% more than non-participants
    c. The average earnings of participants are $1,790
    d. The average treatment effect: participants earn about $1,790 more on average than non-participants

21. Under SLR.1–SLR.5, $\text{Var}(\hat{\beta}_1) = \sigma^2 / \text{SST}_x$. Which change makes $\hat{\beta}_1$ more precise (smaller variance)?
    a. More sample variation in x
    b. Fewer observations
    c. A larger error variance $\sigma^2$
    d. Replacing homoskedasticity with heteroskedasticity

22. In $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u$, the OLS estimate $\hat{\beta}_1$ can also be obtained by:
    a. Regressing y on $x_2$ first, then adding $x_1$
    b. Regressing $x_1$ on $x_2$ (with a constant), then regressing y on the residuals from that regression
    c. Regressing $x_2$ on the residuals from a regression of y on $x_1$
    d. Averaging the slopes of the two simple regressions

23. Which regression violates MLR.3 (no perfect collinearity)?
    a. log(salary) on log(sales) and ceoten
    b. wage on educ and $\text{educ}^2$
    c. wage on educ and exper, where their correlation is 0.9
    d. wage on an intercept, male (= 1 if male) and female (= 1 if female)

24. The true model is $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u$, but $x_2$ is omitted and y is regressed on $x_1$ only. In which case is the simple regression slope unbiased for $\beta_1$?
    a. When the sample size is very large
    b. When $\beta_2 = 0$ or when $x_1$ and $x_2$ are uncorrelated
    c. When the R-squared is high
    d. When $x_2$ is positively correlated with $x_1$ and $\beta_2 > 0$

25. Using 506 communities, the simple regression gives log(price) = 11.71 − 1.043 log(nox), while adding rooms gives log(price) = 9.23 − 0.718 log(nox) + 0.306 rooms. rooms raises price and is negatively correlated with log(nox). Which statement is correct?
    a. The simple regression is biased upward (toward zero), so −1.043 understates the true effect
    b. Adding rooms must always make the nox elasticity more negative
    c. The simple-regression elasticity −1.043 is biased downward (too negative), as predicted by $\beta_2 > 0$ and Corr(log(nox), rooms) < 0
    d. The two estimates differ only because of sampling error

26. High (but not perfect) correlation between $x_1$ and $x_2$ in a multiple regression mainly:
    a. Violates MLR.3, so OLS cannot be computed
    b. Increases the sampling variances of $\hat{\beta}_1$ and $\hat{\beta}_2$ but does not cause bias
    c. Causes $\hat{\beta}_1$ and $\hat{\beta}_2$ to be biased
    d. Inflates the variance of every coefficient, including an unrelated $x_3$

27. The Gauss–Markov theorem states that under MLR.1–MLR.5, the OLS estimators are:
    a. Exactly normally distributed in any sample
    b. The only unbiased estimators of the coefficients
    c. The best estimators even when the errors are heteroskedastic
    d. BLUE: they have the smallest variance among all linear unbiased estimators

28. In a regression, $\hat{\beta}_j = 0.80$ with $se(\hat{\beta}_j) = 0.10$. What is the t statistic for $H_0$: $\beta_j = 1$?
    a. 8.0
    b. −2.0
    c. 2.0
    d. −0.2

29. In a two-sided t test, the p-value is 0.07. Which statement is correct?
    a. $H_0$ is rejected at the 10% level but not at the 5% level
    b. $H_0$ is rejected at the 5% level
    c. $H_0$ is rejected at the 1% level
    d. The probability that $H_0$ is true is 7%

30. The 95% confidence interval for $\beta_1$ is [0.02, 0.10]. Which statement is correct?
    a. In repeated samples, intervals constructed this way contain the true $\beta_1$ 95% of the time, and $H_0$: $\beta_1 = 0$ is rejected at the 5% level (two-sided)
    b. There is a 95% probability that $\hat{\beta}_1$ lies in [0.02, 0.10]
    c. $\beta_1$ equals 0.06 with 95% certainty
    d. $H_0$: $\beta_1 = 0$ cannot be rejected at the 5% level

31. To test $H_0$: $\beta_3 = 0$, $\beta_4 = 0$, $\beta_5 = 0$ jointly, which statement is correct?
    a. The F statistic can be negative when the restrictions help the fit
    b. Three separate t tests always give the same conclusion as the F test
    c. The restricted model always has a smaller SSR than the unrestricted model
    d. Use an F test comparing the SSR of the restricted and unrestricted models; F is large when the SSR rises a lot after imposing the restrictions

32. Which can cause the usual OLS t statistics to be invalid (not t-distributed under $H_0$)?
   a. A sample correlation of 0.95 between two explanatory variables in the model
   b. A low R-squared
   c. Heteroskedasticity, or omitting an important explanatory variable
   d. Having 500 observations instead of 5,000

33. For 32 chemical firms, rdintens = 0.472 + 0.321 log(sales) + 0.050 profmarg, where rdintens is R&D as a percentage of sales. If sales increase by 10%, rdintens changes by about:
   a. 3.21 percentage points
   b. 0.032 percentage points
   c. 32.1 percent
   d. 0.321 percentage points

34. Which statement about the consistency of OLS is correct?
   a. Consistency requires the errors to be normally distributed (MLR.6)
   b. Consistency requires homoskedasticity (MLR.5)
   c. A consistent estimator is unbiased in every finite sample
   d. Under MLR.1–MLR.4, the distribution of $\hat{\beta}_j$ collapses to the true $\beta_j$ as $n \to \infty$, i.e. plim $\hat{\beta}_j = \beta_j$

35. pctstck = $\beta_0 + \beta_1$ funds + $\beta_2$ risktol + u satisfies MLR.1–MLR.4 (pctstck = % of pension in stocks, funds = number of mutual funds offered, risktol = risk tolerance). If funds and risktol are positively correlated and pctstck is regressed on funds only, the inconsistency in $\tilde{\beta}_1$ is:
   a. Zero, because the full model satisfies MLR.1–MLR.4
   b. Negative, because risktol is omitted
   c. It vanishes as the sample size grows
   d. Positive: plim $\tilde{\beta}_1 - \beta_1 = \beta_2\delta_1 > 0$, since $\beta_2 > 0$ and funds and risktol are positively correlated

36. cigs is the average number of cigarettes smoked per day. Can cigs have a normal distribution in the adult population, and what does this mean for OLS?
   a. Yes, by the central limit theorem every variable is normal
   b. No: cigs is nonnegative, piles up at zero (most adults do not smoke) and is discrete; still, with a large n, OLS t and F tests are approximately valid
   c. Yes, because cigs is an average
   d. No, so OLS estimates in a cigs equation are biased

37. Under MLR.1–MLR.5 but without normality (MLR.6), with a large sample:
   a. t tests become invalid no matter how large n is
   b. OLS becomes biased
   c. The usual t and F statistics are approximately valid, because $(\hat{\beta}_j - \beta_j)/\text{se}(\hat{\beta}_j)$ is approximately standard normal
   d. OLS standard errors cannot be computed

38. Under MLR.1–MLR.5, $\mathrm{se}(\hat{\beta}_j) \approx c_j / \sqrt{\{n\}}$. If the sample size is quadrupled, the standard error is approximately:

a. Quartered
b. Halved
c. Unchanged
d. Doubled

39. Crime model (n = 2,725): narr86 is regressed on pcnv, ptime86, qemp86 (restricted model). Its residuals are regressed on pcnv, avgsen, tottime, ptime86, qemp86, giving $R^2 = 0.0015$. To test $H_0$: avgsen and tottime have no effect:

a. $LM = nR^2 \approx 4.09$; compared with $\chi^2_2$ (10% critical value 4.61), $H_0$ is not rejected at the 10% level
b. $LM = 0.0015$, so $H_0$ is rejected at the 1% level
c. $LM \approx 4.09$; compared with $\chi^2_5$, $H_0$ is rejected
d. $LM = n(1 - R^2) \approx 2{,}721$, so $H_0$ is rejected

40. The dependent variable is changed from dollars to thousands of dollars (divided by 1,000). What happens?

a. Only the intercept changes
b. t statistics are divided by 1,000
c. R-squared falls
d. All coefficients and standard errors are divided by 1,000; t statistics and R-squared are unchanged

41. For 32 chemical firms, rdintens = 2.613 + 0.00030 sales − 0.0000000070 sales$^2$ (sales in $ millions). With salesbil = sales/1,000, the coefficients on salesbil and salesbil$^2$ become:

a. 0.00000030 and −0.0000000000070
b. 0.30 and −0.0000070
c. 0.30 and −0.0070, with the intercept and R-squared unchanged
d. 0.030 and −0.00070

42. For rdintens = 2.613 + 0.00030 sales − 0.0000000070 sales$^2$ (sales in $ millions), at what level of sales does the marginal effect of sales on rdintens become negative?

a. About 2,143
b. About 21,429, i.e. $21.4 billion
c. About 42,857
d. At sales = 0

43. A housing equation with dependent variable log(price) contains −0.545 rooms + 0.062 rooms$^2$ (other controls included). Moving from 6 to 7 rooms raises price by about:

a. 6.2%
b. 12.4%
c. −54.5%
d. 19.9%

44. In log(price) = 9.23 − 0.718 log(nox) + 0.306 rooms, what is the exact percentage change in price from one more room?

a. Exactly 30.6%

b. 0.306%

c. 3.06%

d. About 35.8%, = 100[exp(0.306) − 1]

45. In a regression where price and all explanatory variables are standardized (z-scores), the coefficient on nox is −0.340. Interpretation:

a. A 1 ppm increase in nox lowers price by 34%

b. A one standard deviation increase in nox lowers price by 0.34 standard deviation, holding the other variables fixed

c. A 1% increase in nox lowers price by 0.34%

d. nox explains 34% of the variation in price

46. log(wage) = 5.65 + 0.047educ + 0.00078educ· pareduc + 0.019exper + 0.010tenure (n = 722), where pareduc is total years of both parents' education. What is the estimated return to education when pareduc = 32?

a. 4.7%

b. 0.078%

c. About 7.2%

d. 2.5%

47. Which statement about adjusted R-squared is correct?

a. It always rises when a variable is added

b. It is always larger than R-squared

c. It can compare a model for y with a model for log(y)

d. It can fall when a variable is added; it rises only if the new variable's |t| exceeds 1

48. College GPA equation (n = 4,137): predicted colgpa = 2.70 at sat = 1200, hsperc = 30, hsize = 5, with se($\hat{y}^0$) = 0.020 and $\hat{\sigma}$ = 0.560. Which is correct?

a. The prediction interval for one student is narrower than the confidence interval for the mean

b. Both 95% intervals are 2.66 to 2.74

c. The prediction interval shrinks to zero width as n grows

d. The 95% CI for expected colgpa is about 2.66–2.74, but the 95% prediction interval for one student is about 1.60–3.80, because it also includes the error variance $\sigma^2$

49. The dependent variable is log(y). Using exp(predicted log(y)) as the prediction of y:

a. Systematically underestimates E(y|x); it should be scaled up by $\hat{\alpha}_0$, e.g. exp($\hat{\sigma}^2$/2) under normality or the average of exp($\hat{u}_i$)

b. Gives an unbiased prediction of y

c. Overestimates E(y|x)

d. Is valid only when R-squared is high

50. Palm oil production is estimated as: production = 243.63 − 3.34 tax − 0.61 fuel − 0.03 fuel$^2$. Production is the amount of palm oil produced, tax is the tax rate (%), and fuel is the fuel price ($). What is the predicted amount of palm oil produced when the tax rate is 5% and the fuel price is $7?

a. 222.45
b. 221.19
c. 226.93
d. 222.66

51. The palm oil equation above is estimated with an intercept and three regressors (tax, fuel, fuel$^2$). What are the degrees of freedom when the sample size is 85?

a. 84
b. 82
c. 81
d. 80

52. Worker productivity is estimated as: productivity = 3.36 + 0.06 log(training) + 2.53 log(experience) + 0.03 log(training) × log(experience). Productivity is shoes made per hour, training is training hours, experience is years of experience (log = natural log). What is the predicted number of shoes the worker makes in one hour with 2000 hours of training and 17 years of experience?

a. 6.79
b. 10.98
c. 11.63
d. 11.30

53. Vaccination is estimated as: log(vaccination) = 0.34 + 0.15 reminder + 0.23 log(ambassador). Reminder is the number of reminders sent and ambassador is the number of ambassadors who are information hubs (log = natural log). What is the predicted amount of log(vaccination) when 1 reminder is sent and there are five ambassadors?

a. 0.65
b. 0.86
c. 1.64
d. 0.49

54. Vaccination is estimated as: log(vaccination) = 0.34 + 0.15 reminder + 0.23 log(ambassador). Reminder is the number of reminders sent and ambassador is the number of ambassadors who are information hubs (log = natural log). What is the predicted amount of vaccination when 1 reminder is sent and there are five ambassadors?

a. 4.47
b. 0.86
c. 2.36
d. 5.16

# Econometrics Midterm — Answers

**1. c**
A sample is the subset of the population chosen for analysis, and we study it to learn what the population is like. An estimate is a judgment about the population based on the sample.

**2. d**
External validity asks whether a study's conclusions can validly be applied outside the context of that study. It is directly tied to whether the sample represents the population.

**3. b**
Students at one university are unlikely to represent all Korean adults. We must ask whether the sample represents the population, and if it does not, the conclusions are hard to apply to that population (weak external validity).

**4. c**
In multiple regression $\beta_1$ is not a simple slope but a partial slope: the effect with the other regressor (log(sunlight)) held fixed. The dependent variable is log(price), so the units are also in terms of log(price).

**5. b**
log(price) is a linear function of subsidy, so that relationship is linear. But price = $\exp(\beta_0 + \beta_1$ subsidy + ...), so price and subsidy are not linearly related.

**6. c**
**Log-log form:** log(sunlight) and log(price) are linearly related, and $\beta_2$ is an elasticity (a 1% increase leads to about a $\beta_2$% change). Sunlight and price themselves are not linearly related.

**7. d**
$\Delta$ log(price)/$\Delta$ battery $\approx \beta_1 + 2\beta_2$ battery depends on the level of battery, so the relationship is not linear. The model is still linear in the parameters, so OLS can estimate it.

**8. d**
Because battery and sunlight interact, the effect of battery is $\beta_1 + \beta_3 \cdot$ sunlight and depends on the level of sunlight. So $\beta_1$ alone is not the slope of battery, and battery and log(price) are not linearly related.

**9. c**
Random sampling means every element of the population has the same probability of being drawn into the sample. Equal probability of group assignment is random assignment.

**10. d**
Drawing the sample is random sampling. Splitting the drawn people into treatment and control groups by a coin flip is random assignment.

**11. b**
If ability is left out of the model it goes into the error u. If ability and education are correlated, E(u|educ) = 0 (zero conditional mean) fails, so $\beta_1$ is biased even under random sampling.

**12. d**
**Omitted variable bias** = (effect of ability $\beta_2$) × (direction of the correlation between educ and abil). Both are positive, so the bias is positive, that is, the estimate is overstated.

**13. c**
With random assignment, education is not systematically related to factors in the error such as ability, so E(u|educ) = 0 holds and $\beta_1$ is unbiased. (Representativeness is the job of random sampling.)

**14. c**
Observing the same units (cities) repeatedly over time gives panel data. It lets us control for unobserved characteristics that do not change over time.

**15. d**
Combining two cross sections drawn independently at two points in time gives a pooled cross section. The same households are not followed, so it is not a panel. It is often used to compare before and after a policy change.

**16. a**
In a time series the observations are serially correlated and their order matters (trend, seasonality). In a randomly sampled cross section the observations are roughly independent and their order carries no information.

**17. a**
OLS (ordinary least squares) minimizes the sum of squared residuals $\Sigma\hat{u}_i^2$. As a result the residuals always sum to zero (an algebraic property).

**18. b**
$R^2$ = SSE/SST = 1 − SSR/SST is the proportion of the sample variation in y explained by x. It does not mean causality or the size of the slope.

**19. b**
**Log-level (semi-log) form:** a one-unit increase in x changes y by about $100\beta_1$ %. 0.083 × 100 = 8.3%.

**20. d**
If x is 0/1, the slope is the difference between the two group means. Under random assignment OLS is an unbiased estimator of the treatment effect. The non-participant mean is $4,550, so 1,790 is about a 39.3% increase.

**21. a**
The variance rises with the error variance $\sigma^2$ and falls with the sample variation in x, $SST_x$.

**22. b**
**Partialling out:** regress y on the residual of $x_1$ after removing the part explained by $x_2$ (the part uncorrelated with $x_2$), and the slope is $\beta_1$.

**23. d**
With an intercept, male + female = 1 is an exact linear relationship, so there is perfect collinearity. $educ^2$ is not a linear function of educ, and a high correlation (0.9) is multicollinearity, not a violation of MLR.3.

**24. b**
**Omitted variable bias** = $\beta_2 \times \delta_1$. If the omitted variable is irrelevant ($\beta_2 = 0$) or uncorrelated with $x_1$ ($\delta_1 = 0$), there is no bias.

**25. c**
Sign of the omitted variable bias = (effect of rooms $\beta_2$, +) × (correlation between log(nox) and rooms, −) = negative, so the bias is downward. Indeed −1.043 is more negative than −0.718. (Wooldridge problem, HPRICE2)

**26. b**
Multicollinearity only inflates the variances of the related variables (diagnosed with VIF) and unbiasedness is preserved. It is not a violation of MLR.3 in the strict sense.

**27. d**
Under MLR.1–5, OLS has the smallest variance among linear unbiased estimators (BLUE). With heteroskedasticity a better estimator exists. Normality requires MLR.6.

**28. b**
$t = (\hat{\beta}_j - a_j)/se(\hat{\beta}_j) = (0.80 - 1)/0.10 = -2.0$. Subtract the hypothesized value and divide by the standard error.

**29. a**
The p-value is the smallest significance level at which $H_0$ can be rejected. Reject only when $p < \alpha$: $0.07 < 0.10$ but $0.07 > 0.05$.

**30. a**
The confidence interval's endpoints are random (the parameter is not). Zero lies outside the 95% interval, so the two-sided 5% test rejects $\beta_1 = 0$.

**31. d**
$F = [(SSR_r - SSR_{ur})/q] \,/\, [SSR_{ur}/(n-k-1)]$. Imposing restrictions can only raise SSR, so $F \geq 0$, and the individual t tests and the joint test can give different conclusions.

**32. c**
Heteroskedasticity violates MLR.5 and omitting an important variable violates MLR.4 (zero conditional mean), so the t distribution does not hold. A high correlation (0.95) between regressors is not a CLM violation (only perfect collinearity is ruled out). (Wooldridge Problem 4.1)

**33. b**
**Level-log form:** $\Delta y \approx (\beta_1/100)(\%\Delta x) = (0.321/100) \times 10 \approx 0.032$. rdintens is measured in percent, so this is a percentage-point change. (Wooldridge problem, RDCHEM)

**34. d**
**Consistency:** as the sample grows the distribution of the estimator concentrates on the true value. MLR.1–MLR.4 suffice, and even the weaker MLR.4′ ($E(u) = 0$, $Cov(x_j, u) = 0$) is enough. Normality and homoskedasticity are not needed.

**35. d**
**Inconsistency** = $\beta_2 \times \delta_1$ with $\delta_1 = Cov(funds, risktol)/Var(funds)$. A higher risk tolerance means a larger stock share, so $\beta_2 > 0$, and the correlation is positive, giving a positive inconsistency. It does not vanish as the sample grows. (Wooldridge Problem 5.2)

**36. b**
The normal distribution is continuous and symmetric and can take negative values. cigs has a mass at zero, so it is not normal. But normality (MLR.6) is unrelated to unbiasedness, and in large samples inference is approximately valid thanks to asymptotic normality. (Wooldridge Problem 5.3)

**37. c**
Asymptotic normality (central limit theorem): even without normality, t and F tests and confidence intervals are approximately valid in large samples. Homoskedasticity (MLR.5) is still needed, though. With heteroskedasticity the usual standard errors are wrong even in large samples.

**38. b**
Standard errors shrink at the rate $1/\sqrt{n}$. Since $\sqrt{4} = 2$, they fall to about half.

**39. a**
**LM** = $n \cdot R^2_u = 2{,}725 \times 0.0015 \approx 4.09$, the degrees of freedom are the number of restrictions q = 2, and the p-value ≈ 0.13. Regress the restricted-model residuals on all of the regressors. (Wooldridge Example 5.3)

**40. d**
Changing the units of y changes the coefficients and standard errors by the same factor. So the t statistics, p-values and $R^2$ do not change (same statistical conclusions).

**41. c**
**salesbil coefficient** = 0.00030 × 1,000 = 0.30, and salesbil$^2$ = sales$^2$/1,000,000, so its coefficient = −0.0000000070 × $10^6$ = −0.0070. This gives a form that is easy to report. (Wooldridge Problem 6.3)

**42. b**
**Turning point x^*** = $\hat{\beta}_1/(2|\hat{\beta}_2|)$ = 0.00030/(2 × 0.0000000070) ≈ 21,428.57 ($ millions). Beyond it the marginal effect becomes negative. (Wooldridge Problem 6.3)

**43. d**
%Δ price ≈ 100[−0.545 + 2(0.062)(6)] ≈ 19.9%. The turning point is 0.545/(2 × 0.062) ≈ 4.4 rooms, and only about 1% of the communities in the sample have fewer rooms than that. (Wooldridge Example 6.2, HPRICE2)

**44. d**
The 100β approximation for log-level is accurate only for small changes. The exact value is 100[exp(βΔ x) − 1] = 100(1.358 − 1) ≈ 35.8%.

**45. b**
A standardized (beta) coefficient shows how many standard deviations y changes when x changes by one standard deviation, so it lets us compare the effects of variables measured in different units. (Wooldridge Example 6.1)

**46. c**
Δ log(wage)/Δ educ = 0.047 + 0.00078 × pareduc = 0.047 + 0.025 ≈ 0.072. The higher the parents' education, the larger the return to education. (Wooldridge Problem 6.4, WAGE2)

**47. d**
**Adjusted $R^2$** = 1 − [SSR/(n−k−1)]/[SST/(n−1)] penalizes through the degrees of freedom. $R^2$ never falls when variables are added, but adjusted $R^2$ can fall. It cannot be compared across different dependent variables (y vs log y).

**48. d**
Confidence interval for the mean: 2.70 ± 1.96(0.020). Prediction interval for one individual: se($\hat{e}^0$) = $\sqrt{\{0.020^2 + 0.560^2\}}$ ≈ 0.560, giving 2.70 ± 1.96(0.560). The error variance does not shrink as n grows. (Wooldridge Examples 6.5–6.6, GPA2)

**49. a**
**E(y|x)** = exp($\sigma^2$/2)·exp(xβ), so using only exp(log ŷ) underpredicts. Multiply by $\hat{\alpha}_0 > 1$ to correct it. (Wooldridge Section 6.4, Example 6.7)

**50. b**
**production** = 243.63 − 3.34(5) − 0.61(7) − 0.03$(7)^2$ = 243.63 − 16.70 − 4.27 − 1.47 = 221.19.

**51. c**
**df** = n − k − 1 = 85 − 3 − 1 = 81.

**52. c**
**ln(2000)** = 7.6009 and ln(17) = 2.8332
**productivity** = 3.36 + 0.06(7.6009) + 2.53(2.8332) + 0.03(7.6009)(2.8332) = 3.36 + 0.4561 + 7.1680 + 0.6460 = 11.63
log base 10 would give 6.79, so log means ln.

**53. b**
**log(vaccination)** = 0.34 + 0.15(1) + 0.23 ln(5) = 0.34 + 0.15 + 0.23(1.6094) = 0.86.

**54. c**
**vaccination** = exp(0.8602) = $e^{0.8602}$ = 2.36.

# Econometrics Midterm

Short Answer · Econometrics 1–3 (Slides), Wooldridge Ch.1–6, Quiz 2 · 37 questions

1. What is the difference between a population and a sample?

2. Why do econometricians usually study a sample instead of the population, and what is the goal of studying the sample?

3. Define external validity and state which property of the sample matters for it.

4. In $\log(\text{price}) = \beta_0 + \beta_1 \text{subsidy} + \beta_2 \log(\text{sunlight}) + u$, why is $\beta_1$ called a partial slope rather than simply "the slope"?

5. In $\log(\text{price}) = \beta_0 + \beta_1 \text{battery} + \beta_2 \text{sunlight} + \beta_3 \text{battery} \times \text{sunlight} + u$ with $\beta_3 \neq 0$, why does battery not have a linear relationship with log(price)?

6. What is the difference between random sampling and random assignment?

7. For Model 2, $\text{wage} = \beta_0 + \beta_1 \text{educ} + u$ (ability not included), explain why the zero conditional mean condition may be violated.

8. What is a control variable, and why is ability added in Model 1, wage = $\beta_0 + \beta_1$ educ + $\beta_2$ abil + u?

9. What does "ceteris paribus" mean, and why is it important in econometrics?

10. What is panel (longitudinal) data, and what is one advantage of it?

11. What does OLS minimize, and what is a residual?

12. What does R-squared measure in a regression?

13. What does the homoskedasticity assumption (SLR.5) state?

14. What does it mean that the OLS estimator $\beta_1$ is unbiased?

15. When does omitting a relevant-looking variable $x_2$ NOT cause bias in the estimated coefficient on $x_1$?

16. What is the main consequence of multicollinearity (high but not perfect correlation among regressors) for OLS?

17. State the Gauss–Markov theorem.

18. Define the p-value of a hypothesis test.

19. Does a statistically significant coefficient always matter in practice? Answer in 1–2 sentences.

20. Interpret a 95% confidence interval for $\beta_j$.

21. Which can make the usual OLS t statistics invalid (no t distribution under $H_0$): (i) heteroskedasticity, (ii) a sample correlation of 0.95 between two regressors in the model, (iii) omitting an important explanatory variable? Explain briefly.

22. What does it mean that an estimator is consistent, and which assumptions are enough for OLS to be consistent?

23. pctstck = $\beta_0 + \beta_1$ funds + $\beta_2$ risktol + u satisfies MLR.1–MLR.4 (risktol = risk tolerance). If funds and risktol are positively correlated and pctstck is regressed on funds only, what is the sign of the inconsistency in $\tilde{\beta_1}$, and why?

24. Can cigs (average cigarettes smoked per day) have a normal distribution in the adult population? What does this imply for OLS inference?

25. How is the LM (n-R-squared) statistic for testing q exclusion restrictions computed, and what distribution does it follow?

26. If the dependent variable is divided by 1,000 (dollars → thousands of dollars), what happens to the OLS coefficients, standard errors, t statistics and R-squared?

27. What is a beta (standardized) coefficient, and why is it useful?

28. For rdintens = 2.613 + 0.00030 sales − 0.0000000070 $\text{sales}^2$ (sales in $ millions), give the turning-point formula for a quadratic and find where the marginal effect of sales becomes negative.

29. In log(wage) = $\beta_0 + \beta_1$ educ + $\beta_2$ educ· pareduc + $\beta_3$ exper + $\beta_4$ tenure + u (pareduc = both parents' education), what is the return to one more year of education, and what does $\beta_2$ measure?

30. Why is adjusted R-squared used, and when does adding a variable increase it?

31. Why is a 95% prediction interval for one individual outcome $y^0$ much wider than the 95% confidence interval for $E(y|x = x^0)$?

32. When the dependent variable is log(y), why is exp(predicted log(y)) not a good prediction of y, and how is it fixed?

33. Palm oil production is estimated as: production = 243.63 − 3.34 tax − 0.61 fuel − 0.03 fuel$^2$. Production is the amount of palm oil produced, tax is the tax rate (%), and fuel is the fuel price ($). What is the predicted amount of palm oil produced when the tax rate is 5% and the fuel price is $7? Write the equation with the numbers substituted (two decimal places).

34. The palm oil equation above is estimated with an intercept and three regressors (tax, fuel, fuel$^2$). What are the degrees of freedom when the sample size is 85? Write the formula and the numbers.

35. Worker productivity is estimated as: productivity = 3.36 + 0.06 log(training) + 2.53 log(experience) + 0.03 log(training) × log(experience). Productivity is shoes made per hour, training is training hours, experience is years of experience (log = natural log). What is the predicted number of shoes the worker makes in one hour with 2000 hours of training and 17 years of experience? Write the equation with the numbers substituted (two decimal places).

36. Vaccination is estimated as: log(vaccination) = 0.34 + 0.15 reminder + 0.23 log(ambassador). Reminder is the number of reminders sent and ambassador is the number of ambassadors who are information hubs (log = natural log). What is the predicted amount of log(vaccination) when 1 reminder is sent and there are five ambassadors? Write the equation with the numbers substituted (two decimal places).

37. Vaccination is estimated as: log(vaccination) = 0.34 + 0.15 reminder + 0.23 log(ambassador). Reminder is the number of reminders sent and ambassador is the number of ambassadors who are information hubs (log = natural log). What is the predicted amount of vaccination when 1 reminder is sent and there are five ambassadors? Write the equation with the numbers substituted (two decimal places).

# Econometrics Midterm — Model answers

**1.** Econ 1
A population is the entire set of individuals, objects or data points we want to study, while a sample is a subset of the population that is selected for analysis.
*Key results: entire · subset · selected*

**2.** Econ 1
Studying the whole population is often too costly or impossible, so we study a sample
the goal is to find out what the population is like from the sample.
*Key results: costly · impossible · population is like*

**3.** Econ 1
External validity is the validity of applying the conclusions of a study outside the context of that study
it depends on whether the sample is representative of the population.
*Key results: outside · conclusions · representative*

**4.** Econ 2
$\beta_1$ is the change in log(price) from a one-unit increase in subsidy while holding the other variables (log(sunlight)) fixed, so it isolates the effect of subsidy alone.
*Key results: holding · other variables · one-unit*

**5.** Econ 2
Because battery and sunlight interact, the effect of battery on log(price) is $\beta_1 + \beta_3 \cdot$sunlight, which depends on the level of sunlight
the slope is not constant.
*Key results: interact · depends · not constant*

**6.** Econ 3
In random sampling every element of the population has an equal probability of being selected
in random assignment every element has an equal probability of being assigned to a group.
*Key results: equal probability · selected · assigned*

**7.** Econ 3
Ability is omitted, so it goes into the error u
because people with better ability get more education, educ is correlated with u, so $E(u|educ) = 0$ fails and $\beta_1$ is estimated with bias.
*Key results: ability · error · correlated*

**8.** Econ 3
A control variable is included to hold another factor fixed so it no longer sits in the error
adding ability helps satisfy the zero conditional mean condition and removes the bias in $\beta_1$.
*Key results: hold · zero conditional mean · bias*

**9.** Ch.1
Ceteris paribus means "other relevant factors being equal"
most economic questions ask for the causal effect of one variable holding all other factors fixed.
*Key results: other factors · equal · causal*

**10.** Ch.1
Panel data follow the same cross-sectional units over time
they can account for time-invariant unobservables and model lagged responses.
*Key results: same · over time · time-invariant*

**11.** Ch.2
OLS chooses the estimates that minimize the sum of squared residuals
a residual is the actual value $y_i$ minus the fitted value $\hat{y}_i$.
*Key results: sum of squared residuals · actual · fitted*

**12.** Ch.2
$R^2$ = SSE/SST is the fraction of the sample variation in y that is explained by the explanatory variable(s).
*Key results: fraction · variation · explained*

**13.** Ch.2
Homoskedasticity means the error u has the same (constant) variance for any value of x: $Var(u|x) = \sigma^2$.
*Key results: variance · error · same*

**14.** Ch.2
**Unbiasedness means $E(\hat{\beta}_1) = \beta_1$**: if sampling and estimation were repeated many times, the estimates would on average equal the true population value, though one estimate may differ.
*Key results: on average · repeated · true*

**15.** Ch.3
There is no omitted variable bias if the omitted variable is irrelevant (its coefficient is zero) or uncorrelated with $x_1$.
*Key results: irrelevant · uncorrelated*

**16.** Ch.3
Multicollinearity increases the sampling variance of the affected slope estimators, but OLS remains unbiased
it can be detected with variance inflation factors.
*Key results: variance · increase · unbiased*

**17.** Ch.3
Under MLR.1–MLR.5, OLS is BLUE (best linear unbiased estimator): it has the smallest variance among all linear unbiased estimators.
*Key results: BLUE · smallest variance · MLR.1*

**18.** Ch.4
The p-value is the smallest significance level at which the null hypothesis would be rejected
we reject $H_0$ when the p-value is below the chosen level.
*Key results: smallest · significance level · reject*

**19.** Ch.4
Not necessarily: statistical significance only shows the effect differs from zero, while economic (practical) significance depends on the magnitude of the coefficient.
*Key results: not necessarily · magnitude · economic*

**20.** Ch.4
In repeated samples, intervals built this way contain the true population coefficient in 95% of cases
the bounds are random, not the parameter.
*Key results: repeated · contain · true*

**21.** Ch.4
Heteroskedasticity (i) and an omitted important variable (iii) violate the CLM assumptions, so the t statistics are invalid
a high correlation of 0.95 (ii) is not a CLM violation (only perfect collinearity is ruled out).
*Key results: heteroskedasticity · omitted · correlation*

**22.** Ch.5
An estimator is consistent if, as the sample size grows, its distribution collapses to the true parameter (plim $\hat{\beta}_j = \beta_j$)
OLS is consistent under MLR.1–MLR.4 (even zero correlation between each $x_j$ and u is enough).
*Key results: sample size · collapse · true · MLR.4*

**23.** Ch.5
The inconsistency is positive: plim $\tilde{\beta}_1 - \beta_1 = \beta_2\delta_1$, with $\beta_2 > 0$ (higher risktol raises pctstck) and $\delta_1 > 0$ because funds and risktol are positively correlated
it does not disappear as n grows.
*Key results: positive · risktol · correlated · not disappear*

**24.** Ch.5
No: most adults smoke zero cigarettes and cigs is nonnegative and discrete, so it is not normal
with a large sample, asymptotic normality still makes the t and F tests approximately valid.
*Key results: zero · nonnegative · not normal · large sample*

**25.** Ch.5
Estimate the restricted model, regress its residuals on all the explanatory variables, and compute LM = $nR^2$ from that auxiliary regression
under $H_0$ it is approximately chi-square with q degrees of freedom.
*Key results: restricted · residuals · nR · chi-square*

**26.** Ch.6
All coefficients and standard errors are divided by 1,000, while the t statistics and R-squared stay unchanged
only the units change, not the statistical conclusions.
*Key results: divided · unchanged · t statistic · R-squared*

**27.** Ch.6
A beta coefficient comes from regressing the standardized y on standardized x's
it is the change in y, in standard deviations, from a one standard deviation increase in $x_j$, so effects measured in different units can be compared.
*Key results: standardized · standard deviation · units · compare*

**28.** Ch.6
**Turning point** $\hat{x}^* = \hat{\beta}_1/(2|\hat{\beta}_2|) = 0.00030/(2 \times 0.0000000070) \approx 21{,}428.57$, i.e. sales of about $21.4 billion beyond that the estimated marginal effect of sales on rdintens is negative.

**29.** Ch.6
The return is $\beta_1 + \beta_2 \cdot$pareduc, so it depends on parents' education
the interaction coefficient $\beta_2$ is how much the return changes for one more year of pareduc (estimate 0.00078: at pareduc = 32 the return is about 7.2%).
*Key results: depends · parents · interaction*

**30.** Ch.6
R-squared never falls when a variable is added, so adjusted R-squared imposes a penalty for extra regressors using degrees of freedom
adding a variable raises it only if that variable's t statistic exceeds one in absolute value.
*Key results: never falls · penalty · degrees of freedom · t statistic*

**31.** Ch.6
The confidence interval reflects only the sampling variance of $\hat{y}^0$, while the prediction interval also includes the variance of the unobserved error for an individual outcome ($\sigma^2$), which does not shrink as n grows (e.g. colgpa: CI 2.66–2.74 vs PI 1.60–3.80).
*Key results: error · variance · individual*

**32.** Ch.6
$\exp(\log \hat{y})$ systematically underestimates $E(y|x)$
**we multiply it by** $\hat{\alpha}_0 = \exp(\sigma^2/2)$ under normal errors, or by the average of $\exp(\hat{u}_i)$ from the residuals.
*Key results: underestimate · multiply · residuals*

**33.** Ch.6
**production** $= 243.63 - 3.34(5) - 0.61(7) - 0.03(7)^2$
$= 243.63 - 16.70 - 4.27 - 1.47 = 221.19$
*Key results: 221.19 · 16.7 · 4.27 · 1.47*

**34.** Ch.3
**df** $= n - k - 1 = 85 - 3 - 1 = 81$
*Key results: 81 · n − k − 1 · 85 − 3 − 1*

**35.** Ch.6
**ln(2000)** $= 7.6009$ and $\ln(17) = 2.8332$
**productivity** $= 3.36 + 0.06(7.6009) + 2.53(2.8332) + 0.03(7.6009)(2.8332)$
$= 3.36 + 0.4561 + 7.1680 + 0.6460 = 11.63$
log base 10 would give 6.79, so log means ln
*Key results: 11.63 · 7.60 · 2.83 · 7.17 · 0.65*

**36.** Ch.6
**log(vaccination)** $= 0.34 + 0.15(1) + 0.23 \ln(5) = 0.34 + 0.15 + 0.23(1.6094) = 0.86$
*Key results: 0.86 · 0.37 · 1.61*

**37.** Ch.6
**vaccination** $= \exp(0.8602) = e^{0.8602} = 2.36$
*Key results: 2.36 · 0.86*

# Econometrics R

R code functions and output interpretation · W1, W2, W5, W6 · 52 questions

1. Which symbols assign a value to a variable in R, and what does an assignment do?

```
x <- 5
y = x + 2
```

2. Which function combines several values into one vector in R, and how does it work?

```
try1 <- c(9, 8, 7, 6)
```

3. What happens when a single number is added to a vector (try1 + 5), and what happens when two vectors of the same length are added?

```
try1 <- c(9, 8, 7, 6)
try2 <- c(5, 4, 3, 2)
try1 + 5
try1 + try2
```

4. In an R data frame, what do the rows and the columns represent?

```
View(df)
dim(df)
```

5. What statistics does summary() report when it is applied to a numeric vector or to a data frame?

```
summary(x)
```

6. When a data set contains extreme outliers, which measure of the center is more appropriate, the mean or the median, and why?

```
mean(x)
median(x)
```

7. Name two statistical features of a data set that you can read from a histogram created with hist().

```
hist(x, breaks = 15)
```

8. How can table() and sort() be combined in R to find the mode of a numeric variable?

```
table(x)
sort(table(x), decreasing = TRUE)
```

9. In name.R, what does read_file("WAGE2_description.txt") combined with string splitting functions accomplish regarding variable names?

10. In the W2 dataset description, lwage is defined as the natural log of wage. What is the primary econometric purpose of using lwage instead of raw wage?

11. In the W2 dataset, variables like married, black, south, and urban are coded as binary indicators (e.g., =1 if married). How are these interpreted in regression?

12. In Lab 2_2, what is the precise statistical function of setting na.strings="." when importing data via read.csv()?

13. What are the functional role and the potential risk of using attach(practice) and detach(practice) in R?

14. When computing sum(competence - mean(competence)), why does the resulting value theoretically equal zero?

15. Why do we take the square root of variance to compute standard deviation instead of using variance directly?

16. When data is positively skewed with outliers, why is the Interquartile Range (IQR) preferred over standard deviation?

17. In Lab 3_2, what is the key benefit of using formula notation (e.g., mean(mpg ~ am, data = mtcars)) from the mosaic package?

18. Why would an analyst use the explicit package syntax base::mean(mpg ~ am, data = mtcars)?

19. In Lab 4_2, what is the functional advantage of using sapply(data$openingHour, to12) over a standard loop?

20. What does the stringr function str_detect(str, fixed(",")) return, and how is it used in data pipelines?

21. Read the code. In the formula perform ~ eye, which variable is the dependent variable and which is the independent variable, and how does plot() use them?

```
plot(perform ~ eye)
```

22. The code draws a scatterplot of perform on eye, the OLS line (blue), a horizontal line at the mean of perform (red) and a vertical line at the mean of eye (green). Where does the blue line pass, and what does this show about OLS?

```
plot(perform ~ eye)
eye_regr = lm(perform ~ eye)
abline(eye_regr, col="blue")
abline(h=mean(perform), col="red")
abline(v=mean(eye), col="green")
```

23. For the simple regression perform ~ skill, the code compares the model R-squared (about 0.046) with the squared correlation. Why does the script round the difference to 16 digits instead of testing with ==, and what equality is being checked?

```
skill_summary = summary(lm(perform ~ skill))
(skill_coef = cor(perform, skill)^2)
round(skill_summary$r.squared - skill_coef, 16)
```

24. The code fits perform on skill and ravens, and the last line of the summary reports F(2, 82) = 3.3 with p = 0.04184. State the null and alternative hypotheses of this F-test.

```
mult_lm = lm(formula = perform ~ skill + ravens)
(mult_summary = summary(mult_lm))
```

25. In the output F(2, 82) = 3.3 for the model perform ~ skill + ravens, what do the numerator degrees of freedom (2) and the denominator degrees of freedom (82) represent?

```
mult_summary = summary(lm(perform ~ skill + ravens))
mult_summary$fstatistic
```

26. The F-test of the model perform ~ skill + ravens gives F(2, 82) = 3.3 and p = 0.04184. At the 5% significance level, what should we conclude about the model as a whole?

```
# F(2,82) = 3.3, p = 0.04184 (alpha = 0.05)
```

27. The fitted equation of perform on skill and ravens is y = -4.355e-16 + 0.1712 * skill + 0.1746 * ravens, and all variables are standardized (mean 0, standard deviation 1). Interpret the coefficient 0.1712 and explain why the intercept is essentially 0.

```
mult_summary$coefficients
```

28. Read the code. It returns 0.3475 for a group whose skill is 0.5 and whose ravens score is 1.5. What does this number represent and how is it obtained from the model?

```
predict(mult_lm, data.frame(skill=0.5, ravens=1.5))
```

29. Read the code. It supplies two new cases, (skill 0.5, ravens 1.5) and (skill 1.5, ravens 0.5), and asks for a 95% confidence interval. What does the output contain for each case, and what population quantity does the interval describe?

```
predict(mult_lm, data.frame(skill=c(0.5,1.5), ravens=c(1.5,0.5)), interval =
"confidence", level = 0.95)
```

30. The code predicts over 10,000 x-values and shades the region between two columns of pred. What do pred[ ,2] and pred[ ,3] contain, and what does the shaded polygon show?

```
pred <- predict(eye_regr, data.frame(eye=points), interval = "confidence", level = 0.95)
polygon(c(rev(points), points), c(rev(pred[ ,3]), pred[ ,2]), col = rgb(0, 0, 0, 0.1), border = NA)
```

31. Suppose the call shown is changed from interval = "confidence" to interval = "prediction". How does the meaning and the width of the interval change?

```
predict(mult_lm, data.frame(skill=0.5, ravens=1.5), interval = "confidence")
```

32. Read the simulation. Each iteration draws a new sample and stores one number in estimator[rowIdx]. What is the number that is stored, and what does hist(estimator) display after 10,000 iterations (true intercept 3, true slope 0.2, sample size 10)?

```
for(rowIdx in 1:sampleCnt) {
  fertilizer = rnorm(sampleSize, 10, 2)
  yield = intercept + slope * fertilizer + rnorm(sampleSize, 0, 1)
  coef = summary(lm(yield ~ fertilizer))$coefficients
  estimator[rowIdx] = coef[2, 1]
}
hist(estimator)
```

33. In the simulation, t[rowIdx] = coef[2,3] and pValue[rowIdx] = coef[2,4] are stored. What do coef[2,3] and coef[2,4] contain?

```
coef = summary(lm(yield ~ fertilizer))$coefficients
t[rowIdx] = coef[2, 3]
pValue[rowIdx] = coef[2, 4]
```

34. Read the code. The true slope is 0.2, but the null hypothesis says the slope is 0. When is the null rejected, and what does the share of "Reject" in table(rejection) estimate? What would it estimate if the true slope were 0?

```
alpha = 0.05
rejection = pValue <= alpha
rejection = ifelse(rejection, "Reject", "Fail to Reject")
table(rejection)
```

35. Read the code. For 20 samples (true slope 2, sample size 12) it records inside. When is inside TRUE, and about how many of the 20 95% intervals do you expect to contain the true slope? Why might the count differ from that?

```
interval = confint(fer_regr)
lowerBound = interval["fertilizer", "2.5 %"]
upperBound = interval["fertilizer", "97.5 %"]
inside = lowerBound <= slope && slope <= upperBound
```

36. Read the code. How does changing the confidence level from the default 0.95 to 0.99 affect the width of the confidence intervals for the coefficients, and why?

```
confint(mult_lm)
confint(mult_lm, level = 0.99)
```

37. The R-squared of perform ~ skill + ravens is about 0.0745, while the single-predictor squared correlations are about 0.0459 (skill) and 0.0469 (ravens). The check round(mult_summary$r.squared - skill_coef, 2) == 0 returns FALSE. What does this say about adding ravens, and is R-squared the sum of the two single values?

```
mult_summary$r.squared
round(mult_summary$r.squared - skill_coef, 2) == 0
round(mult_summary$r.squared - ravens_coef, 2) == 0
```

38. Read the code from Lab2.R. Why is the first result always 0 apart from rounding error, why are the deviations squared in the second line, and why does the third line divide by n - 1?

```
sum(competence - mean(competence))
sum((competence - mean(competence))^2)
sum((competence - mean(competence))^2) / (length(competence) - 1)
```

39. For the variable grade, quantile(grade) gives a first quartile of 72 and a third quartile of 84. What does IQR(grade) compute, what do the box and whiskers of boxplot(grade) show, and why use the IQR when a distribution is skewed or has outliers?

```
quantile(grade)
IQR(grade)
boxplot(grade)
abline(h=median(grade), col="red")
```

40. For the practice data, the correlation of grade with competence is about 0.83 and with breaks about -0.85. Describe the general pattern each scatterplot shows.

```
plot(grade ~ competence)
plot(grade ~ breaks)
```

41. Read the function to12 from Lab4.R. What does to12("22") return and how? What happens for to12("12"), and why is that a flaw?

```
to12 <- function(str){
  hour = as.integer(str)
  if (hour > 12) {
    return(paste(hour - 12, "PM"))
  } else {
    return(paste(hour, "AM"))
  }
}
```

42. Read the code from Lab4.R. Why is sapply() used to convert the whole column, instead of calling to12(data$openingHour) directly?

```
(openingHour <- sapply(data$openingHour, to12))
(closingHour <- sapply(data$closingHour, to12))
```

43. Explain the role of the lm() function in R and how the formula y ~ x1 + x2 specifies the dependent variable and the independent variables.

```
model <- lm(y ~ x1 + x2, data = df)
summary(model)
```

44. Why is the as.factor() function used on a categorical variable such as session before running a regression in R?

```
model <- lm(y ~ x + as.factor(session), data = df)
```

45. Explain the difference between using "+", ":" and "*" between two variables x1 and x2 in an R model formula.

```
lm(y ~ x1 + x2)
lm(y ~ x1:x2)
lm(y ~ x1*x2)
```

46. What does a significant F-statistic of the overall regression, shown in the last line of summary(model), indicate about the relationship between the dependent variable and the independent variables?

```
summary(model)
```

47. How do you interpret the estimated coefficient of a continuous independent variable in a linear regression model?

```
summary(model)$coefficients
```

48. What is the purpose of the linearHypothesis() function from the car package in R?

```
library(car)
linearHypothesis(model, c("x1 = 0", "x2 = 0"))
```

49. In the regression lm(wage ~ female + educ), female is a dummy variable equal to 1 for women and 0 for men. How is the coefficient of female interpreted?

```
lm(wage ~ female + educ, data = df)
```

50. In the regression lm(wage ~ educ*female), what does the coefficient of the interaction term educ:female measure, and what does the coefficient of female alone measure?

```
lm(wage ~ educ*female, data = df)
```

51. The code compares a restricted model, with one regression for both groups, and an unrestricted model with a full set of interactions with the dummy female. What hypothesis does anova(restricted, unrestricted) test, and what assumption does this test make?

```
restricted <- lm(wage ~ educ + exper, data = df)
unrestricted <- lm(wage ~ (educ + exper)*female, data = df)
anova(restricted, unrestricted)
```

52. In the code, the dependent variable inlf is binary (1 = in the labor force, 0 = not). What do the coefficients of this lm() regression measure, and what is a known drawback?

```
lpm <- lm(inlf ~ educ + age, data = df)
```

# Econometrics R — Model answers

**1.** W1 · Variable assignment
`<-` and = are the **assignment** operators in R
The **right-hand side** (a value or the result of a calculation) is evaluated
The result is **stored** in the variable on the left-hand side
*Key results: assign · right-hand side · stored · left-hand side*

**2.** W1 · Vector creation
The function **`c()`** (concatenate) combines several values into **one vector**
`c(9, 8, 7, 6)` joins the four numbers into a single vector
*Key results: c() · one vector · joins*

**3.** W1 · Vector arithmetic
`try1 + 5` adds 5 to **each element** of the vector and gives 14 13 12 11
Two vectors of the same length are added **element by element**, position by position, so `try1 + try2` gives 14 12 10 8
*Key results: each element · element by element*

**4.** W1 · Structure of a data frame
Each **row** is one **observation** (a subject or a case)
Each **column** is one **variable**
`dim(df)` returns the number of rows and the number of columns
*Key results: observation · variable*

**5.** W1 · The summary() function
`summary()` reports the **minimum** and the **maximum**
It reports the **mean** and the **median**
It reports the first and third **quartiles**
For a data frame it does this for every column
*Key results: minimum · mean · quartile*

**6.** W1 · Mean versus median with outliers
The **median** is more appropriate
The mean is **sensitive** to outliers because every value enters the calculation, so one extreme value pulls it away from the center
The median depends only on the middle position of the sorted data, so it is hardly affected
*Key results: median · sensitive*

**7.** W1 · Reading a histogram
A histogram shows the **shape of the distribution**, for example skewness
It shows the **frequency** (count) of observations in each bin
It shows where the **center** of the data lies
`breaks` controls the number of bins
*Key results: distribution · frequency · center*

**8.** W1 · Finding the mode with table() and sort()
`table(x)` builds a **frequency table** of each value
`sort(table(x), decreasing = TRUE)` orders it in **descending** order
The value with the **highest frequency**, the first entry, is the mode
*Key results: table() · sort() · highest frequency*

**9.** W2 · Reading Description File and Extracting Names
The code reads the dataset description file and performs **string split** and **subsetting** operations to automatically **extract variable names** from WAGE2_description.txt
*Key results: extract variable names · string split*

**10.** W2 · Interpreting Log Wage in W2
Using **lwage** (the **natural log** of wage) linearizes multiplicative relationships and allows coefficients to be interpreted approximately as a **percentage change**
*Key results: natural log · percentage change*

**11.** W2 · Binary Dummy Variables in W2
Variables such as married and urban function as **binary dummy variables**, representing group membership whose coefficients measure the **difference in intercept** relative to the base category
*Key results: binary dummy variables · difference in intercept*

**12.** W2 · Handling Missing Values with na.strings
Setting **na.strings="."** automatically **replaces missing values** (represented by dots) with **NA**, which **prevents calculation errors** when computing summary statistics
*Key results: replaces missing values · prevents calculation errors*

**13.** W2 · Attaching and Detaching Data Frames
`attach()` allows **direct variable access** without typing the dataset prefix, but introduces the risk of a **name conflict** if identical variable names exist
*Key results: direct variable access · name conflict*

**14.** W2 · Sum of Mean Deviations Identity
The sum of deviations from the mean is always zero because **positive and negative deviations** around the mean perfectly **cancel out**
*Key results: cancel out · positive and negative deviations*

**15.** W2 · Variance vs Standard Deviation Units
Variance results in **squared units** that inflate error interpretation, whereas taking the square root reverts the metric back to the **original unit of measurement**
*Key results: squared units · original unit of measurement*

**16.** W2 · IQR Robustness under Skewness
The **IQR** measures the spread of the **middle 50 percent** of the data, making it **robust to outliers** that distort the mean and standard deviation
*Key results: robust to outliers · middle 50 percent*

**17.** W2 · Mosaic Formula Interface Advantages
The **mosaic package** utilizes **formula notation** (~), enabling clean **group-by calculation** without requiring repetitive manual subsetting or indexing
*Key results: formula notation · group-by calculation*

**18.** W2 · Package Namespace Resolution Operator
The double-colon operator performs **namespace resolution** to explicitly invoke a function, allowing you to **override mask** issues when package function names conflict
*Key results: namespace resolution · override mask*

**19.** W2 · Vectorized Function Application using sapply
Using **`sapply()`** achieves **vectorized application** of custom functions across vector elements, which **avoids explicit loops** and improves execution efficiency
*Key results: vectorized application · avoids explicit loops*

**20.** W2 · String Pattern Detection Logic
`str_detect()` executes **pattern matching** to evaluate whether a substring exists, returning a **boolean** **TRUE FALSE** value utilized in conditional logic
*Key results: boolean TRUE FALSE · pattern matching*

**21.** W5 · Scatterplot formula
In **`perform ~ eye`** the variable on the left, **perform**, is the dependent (outcome) variable and goes on the y-axis
the variable on the right, **eye**, is the independent (predictor) variable and goes on the x-axis
That is why the outcome must be written first
*Key results: perform · eye*

**22.** W5 · OLS line and the two means
The blue OLS line always **passes through** the point where the vertical line at the **mean of x** (eye) meets the horizontal line at the **mean of y** (perform), so the fitted line is anchored at the point of means
*Key results: passes through · mean of x · mean of y*

**23.** W5 · R-squared equals squared correlation
It checks that in a simple linear regression **R-squared equals the correlation squared**
The two numbers are computed by different formulas, so they can differ by **tiny floating-point (rounding) differences**
rounding the difference to 16 digits lets the check ignore that noise
*Key results: r-squared equals correlation squared · small differences*

**24.** W5 · Null and alternative of the F-test
The null hypothesis is that **all coefficients are 0** (the skill and ravens slopes are both 0, $\beta_1 = \beta_2 = 0$)
The alternative is that **at least one** of them is not 0
*Key results: all coefficients are 0 · at least one*

**25.** W5 · Degrees of freedom in the F-statistic
The numerator df, 2, is the **number of predictors** tested (skill and ravens)
The denominator df, 82, is the **residual degrees of freedom**, n - k - 1 = n - 3, which implies n = 85 observations
*Key results: number of predictors · residual degrees of freedom*

**26.** W5 · Conclusion of the F-test
Because the p-value 0.04184 is less than 0.05, we **reject the null hypothesis**
At least one of the coefficients is not 0, so the model explains perform **significantly better than a model with no predictors**
*Key results: reject the null hypothesis · at least one · better than*

**27.** W5 · Reading the coefficients
**Holding ravens constant**, a one **standard deviation** increase in skill is associated with a 0.1712 standard deviation increase in predicted perform
Because all variables are **standardized** (mean 0), the regression plane passes through the point of means, so the intercept is about 0
*Key results: holding ravens constant · standard deviation · standardized*

**28.** W5 · Meaning of predict()
The value is the **predicted value (point estimate)** of perform for skill = 0.5 and ravens = 1.5
It is obtained by **substituting** these values into the estimated **regression equation**: -4.355e-16 + 0.1712 * 0.5 + 0.1746 * 1.5, which is about 0.3475
*Key results: predicted value · substitut*

**29.** W5 · Columns returned with interval = "confidence"
For each case (one row per pair) the output gives **fit**, the predicted value (about 0.3475 and 0.3441), plus the **lower and upper** bounds (lwr, upr) of a 95% confidence interval
The interval expresses how accurate our estimate of the **population average** of perform is at those predictor values
*Key results: fit · lwr · population average*

**30.** W5 · Confidence band code
The columns of pred are fit, lwr and upr, so pred[ ,2] holds the **lower bounds** and pred[ ,3] holds the **upper bounds** of the 95% confidence interval at every x-value
The polygon fills the area between them, drawing a shaded **confidence band** around the regression line
*Key results: lower bound · upper bound · confidence band*

**31.** W5 · Confidence interval versus prediction interval
The confidence interval covers the **population average** of the dependent variable, whereas the prediction interval covers an **individual new observation** at those predictor values
Because an individual value also includes its own random error, the prediction interval is **wider**
*Key results: individual · wider · population average*

**32.** W5 · Monte Carlo loop
Every iteration draws a **new random sample**, regresses yield on fertilizer and stores the **slope estimate** (coef[2, 1], the Estimate of fertilizer)
The histogram of these 10,000 slopes shows the **sampling distribution** of the OLS slope estimator, centred near the true slope 0.2
*Key results: slope estimate · sampling distribution · new random sample*

**33.** W5 · Columns of the coefficient table
In the coefficient table the columns are Estimate, Std. Error, t value and `Pr(>|t|)`, and row 2 is the **slope** of fertilizer
So coef[2,3] is the **t-statistic** and coef[2,4] is the **p-value** of the slope test
*Key results: t-statistic · p-value · slope*

**34.** W5 · Rejection rule and what the table estimates
The null is rejected when the **p-value is less than or equal to alpha** (0.05)
With a true slope of 0.2 the null is false, so the share of Reject estimates the test's **power**
If the true slope were 0 (a true null), that share would estimate the **Type I error rate**, about alpha
*Key results: p-value less than alpha · power · type I error*

**35.** W5 · Coverage of confidence intervals
inside is TRUE when the **true slope lies within** that sample's interval [lowerBound, upperBound]
Across repeated samples about **95%**, roughly 19 of 20, of the intervals should contain it, but the count of any one run of 20 varies by **chance** (random sampling)
*Key results: true slope inside · 95% · chance*

**36.** W5 · Confidence level and interval width
The 99% intervals are **wider**
To be **more confident** that the interval captures the true parameter, the interval must be larger, which means a less precise estimate
*Key results: wider · more confident*

**37.** W5 · Comparing R-squared values
Adding ravens makes R-squared **larger** than either single-predictor value (the difference rounds to 0.03, so the check is FALSE)
But 0.0745 is smaller than 0.0459 + 0.0469, so R-squared is **not the sum**: the predictors **overlap** and share some of the explained variance
*Key results: larger · not the sum*

**38.** W5 · Deviations from the mean (Lab2.R)
Deviations from the mean always sum to **zero** because positive and negative deviations **cancel**
Squaring removes the signs so the deviations can be summed, and dividing by **n - 1** gives the **sample variance** (`var(competence)`)
its square root is `sd()`
*Key results: always 0 · squar · n - 1*

**39.** W5 · IQR and boxplot (Lab2.R)
`IQR(grade)` is the **third quartile minus the first quartile** (84 - 72 = 12), the range of the middle 50% of the data, drawn as the box
Whiskers extend to the most extreme points within **1.5 times** the IQR, and points beyond are plotted as **outliers**
The IQR is robust, unlike the mean and standard deviation, when data are skewed or have outliers
*Key results: third quartile minus first quartile · 1.5 · outlier*

**40.** W5 · Reading the scatterplots (Lab2.R)
Grade rises as **competence** rises, a **positive** relationship (r about 0.83)
Grade falls as the number of **breaks** rises, a **negative** relationship (r about -0.85)
*Key results: positive · negative*

**41.** W5 · Function logic (Lab4.R)
`to12("22")` returns **"10 PM"** because 22 is greater than 12, so the function **subtracts 12** and pastes "PM"
`to12("12")` returns **"12 AM"**: 12 is not greater than 12, so it falls in the else branch and **noon is mislabeled as AM**
*Key results: 10 PM · subtract 12 · 12 AM*

**42.** W5 · Why sapply (Lab4.R)
to12 uses **if**, which is not vectorized and handles only a **single value** at a time
`sapply()` applies the function to **each element** of the column and returns the results as a vector
*Key results: each element · if*

**43.** W6 · lm function syntax
The **`lm()`** function fits a **linear model** (an OLS regression)
In the **formula** `y ~ x1 + x2`, the tilde separates the **dependent variable** y on the left from the **independent variables** x1 and x2 on the right, which are joined by plus signs
*Key results: linear model · dependent variable · independent variables · formula*

**44.** W6 · Factor conversion and dummy variables
Using **`as.factor()`** tells R to treat the variable as a **categorical variable** rather than a number
`lm()` then **automatically** creates **dummy variables** for the categories and leaves one out as the **base (reference) category**
*Key results: categorical variable · dummy variables · automatically · base category*

**45.** W6 · Interaction terms in R formulas
"+" includes only the **main effects** of x1 and x2
":" includes only their **interaction** term (the product x1 * x2)
"*" is shorthand for x1 + x2 + x1:x2, so it includes **both** the main effects and the interaction
*Key results: main effects · interaction · both*

**46.** W6 · F-statistic interpretation
A significant F-statistic indicates the **overall significance** of the model
The null hypothesis that all slope coefficients are 0 is rejected, so **at least one** coefficient of the independent variables is **different from zero**
*Key results: overall significance · at least one · not zero*

**47.** W6 · Coefficients and economic significance
The coefficient measures the expected **marginal effect** on the dependent variable of a **one-unit change** in that independent variable, **holding** all other variables in the model **constant**
*Key results: holding constant · unit change · marginal effect*

**48.** W6 · Linear hypothesis testing
The **`linearHypothesis()`** function **tests** linear hypotheses, that is, linear **restrictions** on the **coefficients** of a fitted regression model, such as several coefficients being 0 at the same time or two coefficients being equal
For an lm object it reports an **F-test**
*Key results: restrictions · test · coefficients · F-test*

**49.** W6 · Coefficient of a dummy variable
The coefficient of female is the **difference** in the expected wage between women and the **base group** (men), **holding educ constant**
*Key results: difference · base group · holding educ constant*

**50.** W6 · Interaction with a dummy variable
The coefficient of `educ:female` is the **difference in slopes:** how much the effect of educ on wage (the **return to education**) for women differs from that for men
The coefficient of female alone shifts the **intercept**, the wage difference at educ = 0
*Key results: difference in slopes · return to education · intercept*

**51.** W6 · Testing differences across groups (Chow test)
The null hypothesis is that the **same regression** holds for both groups, that is, the coefficient of female and all its interaction coefficients are 0
The **F-test** compares the SSR of the restricted and unrestricted models (the Chow test), and it assumes a **constant error variance** across the groups
*Key results: same regression · F-test · constant error variance*

**52.** W6 · Binary dependent variable
This is a **linear probability model**
A coefficient measures the change in the **probability** that inlf = 1 for a one-unit change in the regressor
A drawback is that the fitted values can fall **outside 0 and 1**, for example a negative predicted probability
*Key results: probability · linear probability model · outside 0 and 1*